Hierarchical Shell Manifold Theory

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Abstract

Hierarchical Shell Manifold Theory (HSMT) is presented as a graded operator and relational-dynamics framework whose mathematical consequences are separated from claims about realized physical spacetime. The minimal core is a Hilbert space, a complete orthogonal family of grade projections, and a densely defined reference operator. Under explicit additional assumptions, self-adjoint graded realizations admit exact Schur–Feshbach reduction, memory kernels, spectral diagnostics, operationally defined measurement structures, and finite relational benchmarks. A consolidated non-selection analysis shows that grading does not imply observed dimension, positive evolution does not select an operational time or Hamiltonian, and projection does not identify the physical sector. We then examine an explicitly conjectural candidate for physical selection, the maximal collective-relaxation selection principle (MCRSP-1), which hypothesizes that an independently adaptive fixed-resource network selects the fastest slow collective relaxation. In the weighted n-vertex path model, we prove that the unique spectral-gap maximizer has edge weights $w_i = 6Wi(n-i)/[n(n^2-1)]$, giving the five-site ratio $2:3:3:2$ and positive Laplacian spectrum $\{0.4, 1.2, 2.4, 4.0\}$ for $W = 4$. This is a conditional optimization theorem, not evidence of spontaneous selection in nature. An explicit tensor-product counterexample shows that the optimized grade operator leaves independent internal-generator freedom unresolved. The resulting framework is a mathematically testable program with sharply stated physical incompleteness: no unique fundamental generator, operational time, Lorentzian causal structure, observed dimension, or Standard Model dynamics are derived. We give a falsification protocol that distinguishes programmed emulator verification from genuinely spontaneous physical selection.

Scope and evidential status. This article is a mathematical framework paper, not a demonstration of a completed physical theory. In particular, the path-graph optimization associated with MCRSP-1 is a proved conditional theorem; spontaneous selection of its optimizer by a natural adaptive system is a distinct, presently unverified hypothesis; and its identification with the fundamental HSMT generator is a further, currently unsupported physical correspondence. These three claims are never used interchangeably.

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1 Introduction

A recurring difficulty in foundational physics is the tendency to identify a mathematical object with a physical concept before the identification has itself been derived. An operator may generate a one-parameter evolution without thereby defining operational time. A spectral quantity may exhibit a power law without thereby establishing physical spatial dimension. A projected sector may possess internally consistent effective dynamics without explaining why that sector is the one realized in nature. A propagation equation may carry characteristic information without, by that fact alone, establishing a Lorentzian causal cone.

Hierarchical Shell Manifold Theory (HSMT) was developed to investigate how far one can proceed while postponing these identifications. In its mature form, HSMT is not taken to begin with literal concentric spatial shells. Rather, it begins with a graded mathematical architecture and asks which operator, spectral, diagnostic, measurement, and dynamical structures follow from that architecture before any physical spacetime interpretation is inserted.

The minimal frozen structure is represented by

$$(\mathcal{H}, \{P_N\}, D), \tag{1}$$

where $\mathcal{H}$ is a Hilbert-state arena, P_N are mutually orthogonal grade projectors, and D is a densely defined reference operator. Earlier geometric notation of the form

$$\mathcal{M} = \bigsqcup_N \mathcal{M}_N \tag{2}$$

may be used as a realization of the graded structure, but it is not part of the physical interpretation by definition. In particular, the grade label N is not identified with physical dimension.

The mature program is organized around two questions.

(i) What mathematical structure follows from the adopted graded premises?

(ii) What additional input would be required to identify that structure with physical spacetime, dimension, causality, time, or the realized physical sector?

The first question has now been pursued to a controlled internal boundary. The second remains open.

Within declared model classes, HSMT supports controlled self-adjoint operator realizations, exact Schur–Feshbach reduction, energy-dependent effective operators, memory kernels, spectral inverse problems, scale-dependent diagnostics, POVM-based measurement structure, conserved record observables, and exact finite-dimensional benchmark systems. A parallel audit program has also produced negative information: several apparently natural shortcuts do not follow from the current premise set. Grade is not physical dimension by notation; a generator is not operational time; an elliptic or positive relational operator is not a Lorentzian cone; a mode-count exponent is not automatically physical dimension; and an effective sector does not select the physical world.

These negative results are not treated as peripheral failures. They determine the logical content of the mature theory. HSMT therefore presents both a constructive mathematical framework and a sharply delimited physical boundary.

The aims of this paper are:

(1) to state the frozen minimal architecture;

(2) to present the derived operator, spectral, diagnostic, measurement, and dynamical structure in dependency order;

(3) to collect the no-go, non-selection, and valid-stop results that delimit the framework;

(4) to identify the conditional and empirical steps still required for physical completion.

The resulting posture is deliberately conservative: HSMT is presented as an extensible framework with explicitly testable conditional realizations, while framework-level physical selection remains open.

2 Logical Status, Scope, and Interpretive Firewalls

The distinction between mathematical derivation and physical interpretation is sufficiently central that claim status is made explicit throughout the paper.

Definition 2.1 (Claim-status vocabulary)**.** *A statement is labeled according to one of the following classes.*

AXIOM *An adopted primitive of the framework.*

DEFINITION
A mathematical construction or terminological convention.

THEOREM *A rigorous result under explicit hypotheses.*

DERIVED RESULT
A consequence of stated assumptions inside a declared model class.

NO-GO *A theorem, inequality, or counterexample excluding a stronger conclusion within a stated class.*

NON-SELECTION
The present premises do not determine a desired structure; impossibility under future extensions is not claimed.

VALID STOP
A principled boundary at which neither a proposition nor its negation has been derived.

CONDITIONAL RESULT
A conclusion that follows only after an additional bridge or realization assumption is supplied.

HYPOTHESIS
A motivated but presently unproved proposal.

BENCHMARK
A mathematical, numerical, emulator, circuit-model, or experimental test of a declared realization.

OPEN PROBLEM
A question left unresolved by the present framework.

The central interpretive firewalls are summarized in Table 1.

The methodological rule may be stated compactly:

> *A mathematical object is not considered a derived physical observable merely because it can be assigned the name of one.*

Table 1: Interpretive firewalls of mature HSMT.

Mathematical structure	**Permitted statement**	**Not derived automatically**
Grade N	Hierarchy/sector label	Physical dimension
Evolution parameter τ	Mathematical ordering parameter	Operational physical time
Generator H or D	Evolution/spectral operator	Physical clock or temporal metric
Spectral/scaling exponent	Diagnostic of a specified quantity	Observed spatial dimension
Mode count	Combinatorial scaling	Physical dimension
Propagation structure	Mathematical characteristic information	Lorentzian causal cone
Projected sector P	Reduced mathematical sector	Uniquely realized physical world
POVM outcome	Measurement label	Physical position
Record observable	Conserved information	Arrow of time or irreversible measurement

Three kinds of negative conclusion are also distinguished. A *no-go* result rules out a stronger statement within a specified class. A *non-selection* result establishes only that current premises do not determine the desired structure. A *valid stop* records a principled unresolved endpoint. These categories will be used explicitly in Section 9.

Finally, several notions of closure are kept separate:

- *Foundational closure*: the minimal architecture has been reduced to a frozen core.
- *Branch closure*: a particular derivation branch has reached its legitimate endpoint, positive, negative, partial, or unresolved.
- *Valid stop*: no licensed continuation exists from the present premises, but neither side of the unresolved proposition has been proved.
- *Internal derivational closure*: the presently identified branches from the current premise set have been exhausted to their logical boundary.
- *Physical closure*: the missing physical selector and operational bridge have been supplied. This has *not* been achieved.

3 Minimal HSMT Core

3.1 Graded Hilbert structure

Let $\mathcal{H}$ be a complex separable Hilbert space and let

$$\{P_N\}_{N\in\mathcal{I}} \tag{3}$$

be a countable family of mutually orthogonal projections such that

$$P_N P_M = \delta_{NM} P_N, \qquad \sum_{N\in\mathcal{I}} P_N = I \tag{4}$$

in the strong operator topology on the sector under consideration. Define

$$\mathcal{H}_N = P_N \mathcal{H}, \qquad \mathcal{H} = \bigoplus_{N \in \mathcal{I}} \mathcal{H}_N. \tag{5}$$

The label N is the *grade*. Equation (5) is structural; it contains no assertion that N is physical dimension.

Let D be a densely defined reference operator on $\mathcal{H}$. The minimal frozen data are therefore the triplet (1). Depending on realization, D may enter spectral analysis, effective reduction, or a closed evolution law. Its physical interpretation is not fixed at the foundational level.

3.2 Geometric realizations

A geometric realization may attach to the grade structure a disjoint union such as (2). In that case $\mathcal{M}_N$ is a grade-associated geometric object. The mature formulation does not infer that each $\mathcal{M}_N$ is an observed spacetime manifold or that the index N measures physical dimension.

3.3 Dynamical consistency axiom DCP-0

Where a closed evolution is introduced, HSMT adopts the minimal dynamical axiom:

DCP-0. Closed evolution is norm preserving.

In a self-adjoint generator realization one may write

$$U(\tau) = e^{-\mathrm{i}H\tau}, \qquad \|U(\tau)\psi\| = \|\psi\|. \tag{6}$$

The parameter τ is deliberately distinguished from physical time. DCP-0 does not by itself derive an operational clock, Lorentzian signature, causal structure, quantum ontology, or an arrow of time.

4 Graded Operator Structure and Self-Adjointness

For any operator A whose grade blocks are defined, write

$$A_{NM} = P_N A P_M. \tag{7}$$

Formally,

$$A = \sum_{N,M} A_{NM} = A_{\mathrm{diag}} + A_{\mathrm{hop}}, \tag{8}$$

with

$$A_{\mathrm{diag}} = \sum_N A_{NN}, \qquad A_{\mathrm{hop}} = \sum_{N \neq M} A_{NM}. \tag{9}$$

Nearest-neighbor hopping is an additional model restriction,

$$A_{NM} = 0 \quad \text{for} \quad |N - M| > 1, \tag{10}$$

not a consequence of the abstract grade structure.

Theorem 4.1 (Bounded intergrade perturbation)**.** *Let D_0 be self-adjoint on $\mathrm{Dom}(D_0) \subset \mathcal{H}$ and let T be a bounded self-adjoint operator on $\mathcal{H}$. Then*

$$D = D_0 + T \tag{11}$$

is self-adjoint on $\mathrm{Dom}(D_0)$.

Proof. A bounded self-adjoint T is D_0-bounded with relative bound zero. The Kato–Rellich theorem therefore gives self-adjointness of $D_0 + T$ on $\mathrm{Dom}(D_0)$. □

The theorem is insensitive to whether T is diagonal or purely intergrade. For a nearest-neighbor realization one may write schematically

$$T = \sum_N (T_{N,N+1} + T_{N+1,N}). \tag{12}$$

The theorem establishes operator control for the declared realization; it does not establish a uniquely selected universal physical "Master Operator."

4.1 Gaussian coupling realization

A historically productive realization uses a localized intergrade weight of Gaussian form,

$$w(N - M) \propto \exp\left[-\frac{(N - M)^2}{2\sigma^2}\right]. \tag{13}$$

Within one frozen numerical construction,

$$\sigma_0 = \frac{\sqrt{2}}{4} \tag{14}$$

produced nearest-neighbor dominance. This is retained as a numerical/model-specific result. Neither the Gaussian functional form nor the value (14) is selected by the minimal premise set. Possible generalizations such as $\sigma \mapsto \sigma_N$ or grade-dependent separations remain model hypotheses unless supplied by a new derivation or empirical bridge.

5 Projection, Effective Operators, and Memory

Let P be an orthogonal projection onto a selected mathematical sector and let

$$Q = I - P. \tag{15}$$

For a self-adjoint operator H, define the block operators

$$H_{PP} = PHP, \quad H_{PQ} = PHQ, \quad H_{QP} = QHP, \quad H_{QQ} = QHQ. \tag{16}$$

The exact reduction is standard Schur–Feshbach theory.

Theorem 5.1 (Exact projected resolvent)**.** *Let z be such that $z - H_{QQ}$ is invertible on $Q\mathcal{H}$. Define*

$$F_P(z) = z - H_{PP} - H_{PQ}(z - H_{QQ})^{-1}H_{QP}. \tag{17}$$

Whenever $F_P(z)$ is invertible on $P\mathcal{H}$,

$$P(z - H)^{-1}P = F_P(z)^{-1}. \tag{18}$$

The corresponding exact self-energy is

$$\Sigma_P(z) = H_{PQ}(z - H_{QQ})^{-1}H_{QP}, \tag{19}$$

and one may write

$$H_{\mathrm{eff}}(z) = H_{PP} + \Sigma_P(z). \tag{20}$$

The correction (19) is exact under the resolvent hypotheses; it is not a perturbative ansatz.

5.1 Time-domain memory

For a declared evolution law

$$\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}\tau}\left|\Psi(\tau)\right\rangle = H\left|\Psi(\tau)\right\rangle, \tag{21}$$

write $\psi_P = P\Psi$ and $\psi_Q = Q\Psi$. Solving the Q equation formally gives

$$\psi_Q(\tau) = e^{-\mathrm{i}H_{QQ}(\tau-\tau_0)}\psi_Q(\tau_0) \tag{22}$$

$$-\,\mathrm{i}\int_{\tau_0}^{\tau} e^{-\mathrm{i}H_{QQ}(\tau-s)}H_{QP}\psi_P(s)\,\mathrm{d}s. \tag{23}$$

Substitution yields

$$\mathrm{i}\frac{\mathrm{d}\psi_P}{\mathrm{d}\tau} = H_{PP}\psi_P(\tau) + H_{PQ}e^{-\mathrm{i}H_{QQ}(\tau-\tau_0)}\psi_Q(\tau_0) \tag{24}$$

$$-\,\mathrm{i}\int_{\tau_0}^{\tau} K(\tau-s)\psi_P(s)\,\mathrm{d}s, \tag{25}$$

with exact memory kernel

$$K(\Delta\tau) = H_{PQ}e^{-\mathrm{i}H_{QQ}\Delta\tau}H_{QP}. \tag{26}$$

Memory here is a consequence of eliminating Q degrees of freedom. It is not an independent microscopic arrow of time.

5.2 Isolation estimate

If $H_{QP} = H_{PQ}^{\dagger}$ and z lies away from $\mathrm{spec}(H_{QQ})$, then

$$\|\Sigma_P(z)\| \le \|H_{PQ}\|^2\,\|(z-H_{QQ})^{-1}\| \le \frac{\|H_{PQ}\|^2}{\mathrm{dist}(z,\mathrm{spec}(H_{QQ}))}. \tag{27}$$

This provides a transparent weak-coupling/isolation criterion.

5.3 Projection is not selection

The logical input to Theorem 5.1 includes P. The theorem does not derive P. This distinction is later formalized by PB-0: Feshbach reduction is conditional dynamics given a decomposition, not a selector.

6 Spectral Structure and Inverse Problems

6.1 Scalar Green function and self-energy

In the scalar realization used in Application A0, the visible resolvent is written

$$G(z) = \frac{1}{z-\omega_0-\Sigma(z)}. \tag{28}$$

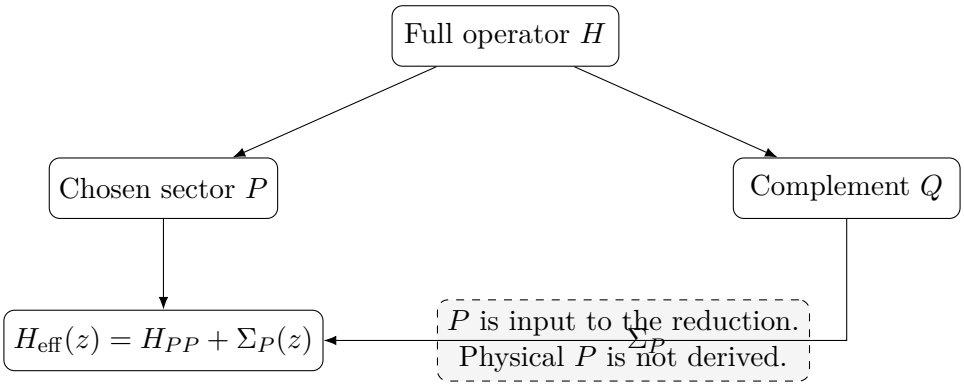


Figure 1: Projection does not equal physical selection. The Schur–Feshbach map exactly reduces a chosen sector; it does not choose that sector.

Whenever $G(z) \neq 0$,

$$\Sigma(z) = z - \omega_0 - G(z)^{-1}. \tag{29}$$

The map (29) is algebraically exact within the declared scalar model.

Let μ be a positive contact spectral measure and define

$$m(z) = \int \frac{\mathrm{d}\mu(M)}{z - M}, \qquad \Sigma(z) = g^2 m(z). \tag{30}$$

For $\operatorname{Im} z > 0$, $\operatorname{Im} m(z) \leq 0$. If $\mathrm{d}\mu(M) = \rho(M)\mathrm{d}M$ locally, then

$$\rho(E) = -\frac{1}{\pi g^2} \lim_{\eta \to 0^+} \operatorname{Im} \Sigma(E + \mathrm{i}\eta). \tag{31}$$

At finite resolution,

$$\rho_\eta(E) = -\frac{1}{\pi g^2} \operatorname{Im} \Sigma(E + \mathrm{i}\eta), \tag{32}$$

which is the Poisson-broadened reconstruction of the underlying density.

6.2 Application A0: semicircle benchmark

The frozen benchmark uses

$$\omega_0 = 0.3, \qquad g^2 = 0.4, \qquad \eta = 0.02, \tag{33}$$

with normalized semicircle contact density

$$\rho_{\mathrm{sc}}(M) = \begin{cases} \dfrac{\sqrt{4 - M^2}}{2\pi}, & |M| \leq 2, \\ 0, & |M| > 2. \end{cases} \tag{34}$$

Its Stieltjes transform is

$$m_{\mathrm{sc}}(z) = \frac{z - \sqrt{z^2 - 4}}{2}, \tag{35}$$

where the branch is chosen so that $\sqrt{z^2 - 4} \sim z$ as $|z| \to \infty$. For $-2 < E < 2$,

$$m_{\mathrm{sc}}(E + \mathrm{i}0) = \frac{E - \mathrm{i}\sqrt{4 - E^2}}{2}, \tag{36}$$

which recovers (34) through (31).

The corresponding memory transform is

$$\int_{-2}^{2} \rho_{\mathrm{sc}}(M) e^{-\mathrm{i}M\tau}\, \mathrm{d}M = \frac{J_1(2\tau)}{\tau}. \tag{37}$$

Since $J_1(2\tau) \sim (\pi\tau)^{-1/2}\cos(2\tau - 3\pi/4)$ up to subleading corrections,

$$\mathrm{memory}(\tau) \sim \tau^{-3/2} \times \mathrm{oscillation}, \tag{38}$$

so the exact asymptotic exponent is $p = 3/2$.

The completed numerical benchmark produced the results in Table 2.

Table 2: Frozen Application A0 benchmark results.

Quantity	**Frozen result**
Noiseless $G \to \Sigma$ maximum absolute error	$\approx 6.66 \times 10^{-16}$
Noiseless $G \to \Sigma$ relative L^2 error	$\approx 3.73 \times 10^{-16}$
Finite-resolution interior density NRMSE	$\approx 0.984\%$
Fitted memory exponent	$p = 1.49805$ versus exact 1.5
Numerical total measure weight	≈ 0.9999992
Off-support poles in declared scan $[-5, 5]$	none found for the declared parameter set
10^{-4} multiplicative complex G noise	Σ relative L^2 error $\approx 3.77 \times 10^{-4}$
10^{-3} multiplicative complex G noise	Σ relative L^2 error $\approx 3.65 \times 10^{-3}$
10^{-2} multiplicative complex G noise	Σ relative L^2 error $\approx 3.62 \times 10^{-2}$

These results validate the numerical inverse engine against the declared analytic benchmark; they do not establish that the semicircle contact measure is physically realized.

6.3 Conditional inverse-square branch

A historical asymptotic branch uses an exponent β_Ω taking the values

$$\beta_\Omega = \frac{5}{6} \qquad \text{or} \qquad \beta_\Omega = \frac{7}{6}. \tag{39}$$

The inherited coefficient map is

$$c(\beta_\Omega) = \frac{\beta_\Omega(2 - \beta_\Omega)}{4}, \tag{40}$$

so both branches give

$$c = \frac{35}{144}, \qquad -\frac{c}{y^2} = -\frac{35}{144y^2}. \tag{41}$$

For the corresponding inverse-square equation, the Bessel-type index is

$$\nu_B = \sqrt{\frac{1}{4} - c} = \frac{1}{12}. \tag{42}$$

This value is mathematically correct within the historical inverse-square branch; it is not independently selected by the minimal HSMT core.

Under the additional historical identifications, the retained conditional chain is

$$\Omega(y) \sim y^{5/6} \text{ or } y^{7/6} \Rightarrow -\frac{35}{144y^2} \Rightarrow \nu_B = \frac{1}{12} \Rightarrow \mathrm{DtN} \sim A^{1/12}$$
$$\Rightarrow M^{-1/6} \Rightarrow \mathrm{d}\mu(M) \sim M^{-1/3}\mathrm{d}M \Rightarrow \Sigma \sim p^2 E^{-1/3} \Rightarrow E \sim p^{3/2}. \quad (43)$$

Equation (43) is retained explicitly as a *conditional historical scaling chain*. The intermediate arrows after the inverse-square step rely on source-specific identifications and are not reconstructed here as new derivations.

6.4 WKB frame-density map

A separate branch begins from a chosen asymptotic superpotential

$$W(y) \sim g y^s, \qquad g > 0. \quad (44)$$

The frozen WKB counting result is

$$\frac{\mathrm{d}n}{\mathrm{d}y} \sim y^s. \quad (45)$$

Defining $\beta_{\mathrm{fr}} = s$, one obtains

$$n(y) \sim C y^{s+1}, \qquad y_n \sim n^{1/(s+1)}. \quad (46)$$

For harmonic confinement, $s = 1$ and therefore

$$y_n \sim n^{1/2}. \quad (47)$$

Proposition 6.1 (WKB density map and non-selection)**.** *Given an inserted asymptotic superpotential $W(y) \sim g y^s$ in the declared WKB regime,* (45)–(46) *follow. The current HSMT premises do not select s or uniquely select W.*

The frame index n is not automatically the grade N. Likewise a coordinate rewrite such as $s = 1/a$ is a reparameterization unless a separate dynamical principle selects a.

7 Dimensional Diagnostics and Mode Counting

Let $Q_N(\ell) > 0$ be a declared scale-dependent quantity associated with grade N. Define

$$d_N(\ell) = \frac{\mathrm{d}\ln Q_N(\ell)}{\mathrm{d}\ln \ell} = \frac{\ell}{Q_N(\ell)} \frac{\mathrm{d}Q_N}{\mathrm{d}\ell}. \quad (48)$$

If $Q_N(\ell) = C\ell^d$ on a scale interval, then $d_N(\ell) = d$ there. If no exact power law holds, d_N is a running logarithmic exponent. A finite-resolution estimator is

$$d_N[\ell_1, \ell_2] = \frac{\ln Q_N(\ell_2) - \ln Q_N(\ell_1)}{\ln \ell_2 - \ln \ell_1}. \quad (49)$$

The mature framework distinguishes

$$d_{\mathrm{geom}}, \qquad d_{\mathrm{res}}, \qquad d_{\mathrm{spec}}, \qquad d_{\mathrm{dyn}}. \quad (50)$$

No equality among the quantities in (50) is assumed or generally derived.

The logical order is therefore

$$N \longrightarrow \text{specify or derive } Q_N(\ell) \longrightarrow d_N(\ell), \tag{51}$$

not $N = d_N$. Neither a monotone progression of d_N across grades nor a fractional interpolation of physical dimension between grades is derived by the current premise set.

7.1 Mode counting and MS-C0

The frozen SRB analysis contains the asymptotic mode counts

$$Q_I(M) \sim \frac{M^3}{6}, \qquad Q_H(M) \sim \frac{M^3}{6} \tag{52}$$

for large M in the declared construction. Therefore

$$\frac{\mathrm{d}\ln Q}{\mathrm{d}\ln M} \longrightarrow 3. \tag{53}$$

No-Go Result 7.1 (MS-C0: mode count does not select physical dimension)**.** *If $Q(M) \sim CM^k$ with $C > 0$, then the logarithmic count exponent tends to k. This establishes scaling with respect to the variable M. It does not establish that physical dimension equals k unless M and Q are independently given the required operational geometric interpretation.*

Accordingly, (52) does not imply three-dimensional physical space. The frozen datum $q_I(4) = 2$ is likewise a structural fact of its combinatorial/operator construction and is not by itself a physical transition or dimension statement.

8 Dynamics, Measurement, Packets, and Records

8.1 POVM structure

Let X be an outcome space. A positive-operator-valued measure E assigns positive operators $E(\Delta) \geq 0$ to measurable sets $\Delta \subseteq X$, satisfies $E(X) = I$, and is countably additive in the weak operator topology. For a state ρ,

$$p_\rho(\Delta) = \mathrm{Tr}[\rho E(\Delta)]. \tag{54}$$

The frozen PC sequence establishes that the relevant internal-fibre HSMT construction admits a canonical POVM. "Canonical" here is relative to that internal structure; it does not mean unique physical detector, observer, spacetime localization, or measurement ontology.

Informational completeness on a state class $\mathcal{S}$ means that equality of all outcome probabilities implies equality of states:

$$\mathrm{Tr}(\rho E_x) = \mathrm{Tr}(\sigma E_x)\ \forall x \quad \Longrightarrow \quad \rho = \sigma, \qquad \rho, \sigma \in \mathcal{S}. \tag{55}$$

The frozen result is informational completeness on the declared internal fibre. This does not make the outcome labels physical positions.

8.2 Packets and localization

Packet-like states may be useful representations or probes. The frozen PC sequence does not, however, select a canonical localized packet. Moreover, OMP-3F closes at a valid stop: packet-frame completeness remains undetermined. Neither completeness nor incompleteness is licensed.

The scalar Q_N used in the measurement layer is a diagnostic only. There is no contradiction between a full internal-fibre POVM being informationally complete and Q_N being only one compressed scalar functional of the measurement structure.

8.3 Record persistence

Let H be self-adjoint and let Q_a be bounded, or satisfy the appropriate strong-commutation/domain conditions. If

$$[H, Q_a] = 0, \tag{56}$$

then

$$e^{\mathrm{i}H\tau} Q_a e^{-\mathrm{i}H\tau} = Q_a \tag{57}$$

and hence

$$\mathrm{Tr}[\rho(\tau) Q_a] = \mathrm{Tr}[\rho(0) Q_a]. \tag{58}$$

This is the frozen RS-0 persistence condition. Conservation of Q_a does not by itself establish a physical clock, an observer, irreversible measurement, or a thermodynamic arrow.

A possible future temporal bridge would require additional operational structure: a monotonic clock variable, reproducible subsystem comparisons, and an independently justified temporal normalization. Those are not supplied by RS-0 itself.

9 No-Go and Non-Selection Results

The mature HSMT record includes explicit no-go theorems, non-selection results, and valid stops. Their logical distinction is important.

9.1 NI-0: monotone-coupling no-go

Consider

$$H(q) = H_* + \sum_e f_e(q_e) O_e, \qquad O_e \geq 0, \qquad f_e'(q_e) > 0. \tag{59}$$

Where the ground-state branch $E_0(q)$ is differentiable, Hellmann–Feynman gives

$$\frac{\partial E_0}{\partial q_e} = f_e'(q_e) \langle O_e \rangle_0 \geq 0. \tag{60}$$

No-Go Result 9.1 (NI-0: monotone positive-coupling no-go)**.** *Within the class* (59)*, ordinary q-independent interactions combined with monotone positive-semidefinite q-dependent couplings cannot by themselves generate the required interior restoring minimum along q.*

This is class-specific. It does not prove that no stabilization mechanism can exist after genuinely new structural input is added.

9.2 NI-1F: Gaussian measured-pullback monotonicity

For finite positive-semidefinite operators A and B under the frozen Gaussian measured-pullback construction, the associated real-positive functional $F(q)$ obeys

$$F'(q) \leq 0, \tag{61}$$

with equality only in the coincident-pullback case.

No-Go Result 9.2 (NI-1F)**.** *The frozen Gaussian measured-pullback response is monotone in q in the declared finite positive-semidefinite class and therefore does not generate an interior canonical vacuum selector through a sign change of $F'(q)$.*

The frozen NI-1 status is partial-positive structural / negative canonical-vacuum / new input required.

9.3 S+C minimal nonfundamental benchmark

The S+C family is retained explicitly as a *minimal nonfundamental benchmark*. An S move is a split/merge operation with

$$\Delta V = +1, \qquad \Delta E = +1, \tag{62}$$

while preserving β_1. A C move is a local cycle activation/deactivation with

$$\Delta V = 0, \qquad \Delta E = \pm 1, \tag{63}$$

and changes β_1. The firewall is

$$r \neq N, \qquad \chi = \kappa_C/\kappa_S \text{ unfixed.} \tag{64}$$

The family diagnoses refinement and cycle structure; it does not select a fundamental microscopic law.

9.4 B3, B5, and B7

The post-PI0 branches close as non-selection results:

- B3: the current premises do not select the proposed $V_{\mathrm{gr-rel}}$ bridge;
- B5: the current premises do not derive a distinguished seed or absolute grade origin;
- B7: the current premises do not select the candidate $W(\rho)$ law.

Each may define a valid model after new input; none is internally selected by the frozen core.

9.5 Transport non-selection

Locality, many-body dephasing, or clustering alone do not imply diffusion. A local unitary system may be ballistic, diffusive, anomalous, or localized depending on conserved quantities and current correlations. A genuine diffusive claim requires, in particular, absence of a nonzero ballistic/Drude component and suitable Green–Kubo integrability, schematically

$$D \propto \int_0^\infty \langle J(\tau)J(0)\rangle_c \, \mathrm{d}\tau \tag{65}$$

after the appropriate conserved-mode projections and normalization.

9.6 Positivity counterexample

The SRB audit distinguishes positive-semidefinite Gram structure, elementwise positivity, and strict positive overlap. These notions are not interchangeable. A frozen S^1 Fourier construction supplies a counterexample to the superseded stronger universal positivity claim. The exact Fourier formula is a source-fidelity item for the appendix; the scientific status is already frozen as a counterexample/no-go.

9.7 OMP-3F valid stop

Boundary Result 9.3 (OMP-3F)**.** *Within the analyzed packet/frame construction, the available premises do not establish either completeness or incompleteness of the candidate packet frame. The branch closes at a valid stop.*

9.8 BEGC no-smuggling comparison

The frozen BEGC-0 principle states:

> "At every physically admissible response level, externally relevant relations are generated from robustly controlled-complexity exposed states by a universal regular response law of robustly controlled nomological complexity. Independent pair-indexed response data are not fundamental inputs."

For

$$W_{ab} = F_\theta(z_a, z_b), \qquad \dim z = r, \qquad \dim \theta = p, \tag{66}$$

with r, p independent of n, the frozen finite-resolution comparison is

$$K_{\text{ext}}^{(\epsilon)} = O\left(nr \log \frac{1}{\epsilon} + p \log \frac{1}{\epsilon}\right) \tag{67}$$

versus generic pairwise data complexity

$$\Theta\left(n^2 \log \frac{1}{\epsilon}\right). \tag{68}$$

Thus the hierarchical family occupies positive codimension in the generic pair-data family for sufficiently large n. This is a conditional mathematical no-smuggling/compressibility result, not a physical dimension or selector theorem.

9.9 PB-0: physical projector principled stop

The frozen status is:

> **PB-0 = CLOSED / CURRENT CORE ANALYZES BUT DOES NOT IDENTIFY PHYSICAL P/Q / EMPIRICAL OR MICROSCOPIC BRIDGE REQUIRED.**

The exact frozen statements include:

> "The current core analyzes but does not identify a physical P/Q decomposition."
>
> "Physical-sector/projector identification is frozen as empirical or new microscopic bridge required."
>
> "Feshbach reduction is conditional dynamics given a decomposition, not a selector."
>
> "No further internal projector-selection search is licensed under the present core."

"Multi-contact response strengthens reconstruction and supplies cross-contact falsification but does not derive the grade decomposition."

"No concrete already-existing HSMT object operationally identifies a physical projector/ contact system rather than merely analyzing a chosen mathematical P."

9.10 PB0-N1 through PB0-N7

The following are reproduced verbatim from the frozen record.

Boundary Result 9.4 (PB0-N1)**.** *THE NUMERICAL VALUE OF c IS NOT AN INVARIANT GRADE SELECTOR. ONLY DIMENSIONLESS PROPAGATION RELATIONS OR THE CHARACTERISTIC / CAUSAL STRUCTURE HAVE AN INVARIANT CLAIM TO PHYSICAL SIGNIFICANCE.*

Boundary Result 9.5 (PB0-N2)**.** *A CHARACTERISTIC-PROPAGATION SELECTOR CAN IDENTIFY A GRADE ONLY TO THE EXTENT THAT THE GRADE-TO-CHARACTERISTIC MAP IS INJECTIVE OR OTHERWISE OPERATIONALLY DISCRIMINATING.*

Boundary Result 9.6 (PB0-N3)**.** *CHARACTERISTIC-PROPAGATION DATA MAY IDENTIFY A PHYSICAL GRADE REGIME WITHOUT FIXING AN ABSOLUTE GRADE ORIGIN.*

Boundary Result 9.7 (PB0-N4)**.** *A POSITIVE RELATIONAL / ELLIPTIC OPERATOR DOES NOT BY ITSELF DERIVE A LORENTZIAN CHARACTERISTIC CONE. PLACING THAT OPERATOR INTO A WAVE EQUATION IS AN ADDITIONAL DYNAMICAL CHOICE.*

Boundary Result 9.8 (PB0-N5)**.** *A POSTULATED GRADE-DEPENDENT SPEED c_N DOES NOT CONSTITUTE A PHYSICAL HSMT THEORY UNLESS THE GRADE DEPENDENCE, SCALE CALIBRATION, AND OBSERVABLE CONSEQUENCES ARE DERIVED INDEPENDENTLY.*

Boundary Result 9.9 (PB0-N6)**.** *A SPATIAL / RELATIONAL PRINCIPAL FORM ALONE CANNOT DETERMINE A PHYSICAL LIGHT SPEED. A RELATIVE NORMALIZATION BETWEEN TEMPORAL AND SPATIAL PRINCIPAL TERMS IS REQUIRED.*

Boundary Result 9.10 (PB0-N7)**.** *CAUSAL-CONE RECOVERY AND DIMENSION RECOVERY ARE DISTINCT PROBLEMS. DERIVING ONE DOES NOT BY ITSELF DERIVE THE OTHER.*

9.11 Global non-selection boundary

The present frozen premises do not derive:

(a) physical spacetime;

(b) observed physical dimension;

(c) operational physical time;

(d) Lorentzian causal structure;

(e) the uniquely realized physical P/Q decomposition;

(f) universal geometry across independently emergent sectors.

These are non-selection statements relative to the present premise set, not impossibility theorems against future extensions.

10 Foundational Closure and HSMT-FM0

The maturation program terminates internally at HSMT-FM0. The terminal statement is:

> "Internal derivation from the current premise set is complete to the presently identified logical boundary."

This explicitly does not mean that the physical theory is complete.

The mature five-stratum organization is:

F0 minimal primitives: $(\mathcal{H}, \{P_N\}, D)$ and, where used, DCP-0;

F1 internal mathematical consequences: operator control, projection, spectral structure, diagnostics, measurement, records, benchmarks;

F2 internal limits: no-go, non-selection, and valid-stop results;

F3 conditional physical bridges;

F4 external physical completion, presently absent.

Boundary Result 10.1 (HSMT-FM0)**.** *Given the currently frozen HSMT premises and completed audited branches, the internal open derivation ledger is empty to the presently identified boundary. Physical spacetime, physical dimension, Lorentzian causal structure, the uniquely realized physical sector, and multi-sector physical universality remain underived. Further physical closure therefore requires genuinely new information not already contained in the closed premise set.*

The status is

HSMT-FM0	FROZEN
Internal derivation	COMPLETE TO BOUNDARY
Physical theory	OPEN
Next legitimate activity	SYNTHESIS, EXPERIMENT, OR EXTERNAL-BRIDGE DISCOVERY

The selector role itself is identified as the MC-1 envelope, but no current result occupies it. PGCH, numerical c, a chosen P, an inserted c_N, a mode-count exponent, or a convenient Gaussian kernel do not fill the envelope merely by being mathematically useful.

11 Conditional Physical Bridges

HSMT presently supports conditional statements of the form

$$\text{if bridge } B \text{ is independently justified, then consequence } C \text{ follows.} \tag{69}$$

The bridge itself is not selected by the current premise set.

Conditional Result 11.1 (Projected-sector bridge)**.** *If a particular mathematical sector P is independently identified as physically realized, then Theorem 5.1 supplies its exact reduced response under the stated operator hypotheses.*

Conditional Result 11.2 (Spectral/geometric bridge)**.** *If a declared spectral diagnostic is operationally shown to encode physical geometry, then its scale dependence may acquire physical geometric meaning.*

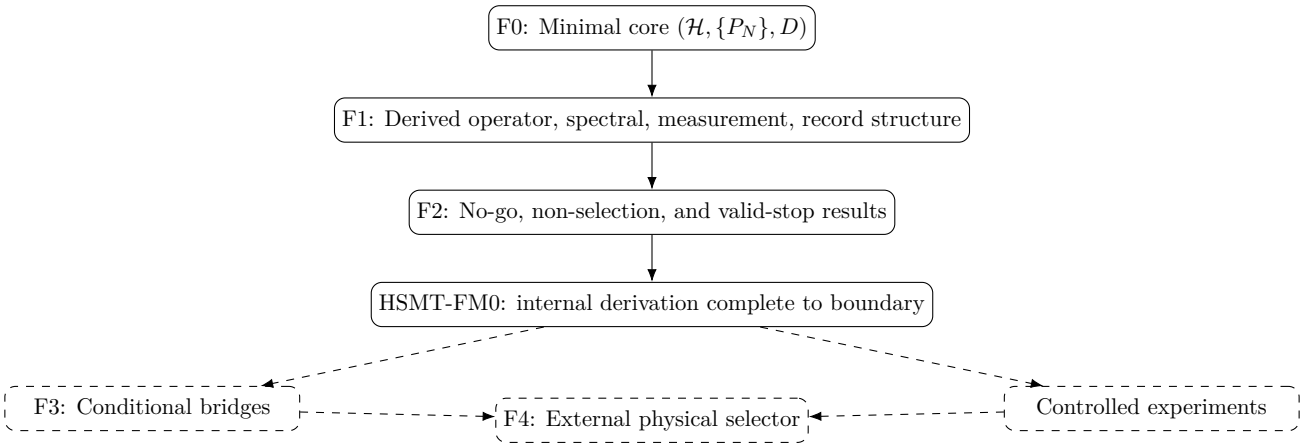


Figure 2: Logical architecture of mature HSMT. The transition from internal closure to physical selection remains open.

Conditional Result 11.3 (Dimensional bridge)**.** *If $Q_N(\ell)$ is connected to an operational physical observable satisfying the required scaling assumptions, then $d_N(\ell)$ may acquire physical dimensional meaning. Equation* (48) *alone does not provide that identification.*

Conditional Result 11.4 (Temporal bridge)**.** *If the mathematical evolution parameter τ is connected consistently to operational clocks and persistent record structure with the required normalization, an emergent physical-time interpretation may become available.*

Conditional Result 11.5 (Causal bridge)**.** *If a mathematical propagation structure is supplemented by an independently justified principle establishing operational signaling constraints and Lorentzian causal properties, it may support a physical causal interpretation.*

Conditional Result 11.6 (Multi-sector universality bridge)**.** *If cross-sector dynamics or empirical constraints force independently emergent sectors to share one effective geometry, their structures may support a common spacetime interpretation.*

PGCH is retained as a legitimate research hypothesis, not as an MC-1 selector. Likewise the historical chain (43) remains conditional.

11.1 Criterion for reopening derivation after FM0

A future foundational branch is legitimate only if it introduces genuinely new information. Qualifying inputs include:

(a) reproducible empirical discrimination among current realizations;

(b) an independently motivated microscopic law constraining grade, projector, generator, or bridge structure;

(c) an external mathematical principle not already encoded in the frozen core;

(d) an independently derived cross-theory identification that supplies an operational bridge without inserting the HSMT conclusion by hand.

Reparameterization, basis choice, preferred-grade declaration, fitted c_N, insertion of a wave equation, or renaming a count exponent as dimension do not qualify.

12 Computational and Hardware Benchmarks

The benchmark program is explicitly separated into levels of evidence. Exact mathematics, numerical verification, emulator realization, SPICE implementation, physical hardware measurement, and physical-theory interpretation are distinct.

12.1 P3 exact benchmark

The minimal path-graph Laplacian is

$$K = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}, \tag{70}$$

with

$$\operatorname{spec}(K) = \{0, 1, 3\}. \tag{71}$$

The positive spectrum is $\{1, 3\}$.

12.2 Six-real-state S and I realizations

Let $u, v \in \mathbb{R}^3$ and $Y = (u, v)$. The S realization is

$$\dot{u} = gKv, \qquad \dot{v} = -gKu. \tag{72}$$

Hence

$$\ddot{u} = -g^2K^2u, \qquad \omega_S(\lambda) = g\lambda. \tag{73}$$

The I realization is

$$\dot{u} = gv, \qquad \dot{v} = -gKu, \tag{74}$$

so

$$\ddot{u} = -g^2Ku, \qquad \omega_I(\lambda) = g\sqrt{\lambda}. \tag{75}$$

Thus the spectral exponent is $\alpha_S = 1$ for S and $\alpha_I = 1/2$ for I.

For the P3 positive modes,

$$\frac{\omega_{S,2}}{\omega_{S,1}} = 3, \qquad \frac{\omega_{I,2}}{\omega_{I,1}} = \sqrt{3}. \tag{76}$$

12.3 P5 controlled discriminator

The four positive P5 path eigenvalues are

$$\lambda_1 = \frac{3 - \sqrt{5}}{2}, \qquad \lambda_2 = \frac{5 - \sqrt{5}}{2}, \tag{77}$$

$$\lambda_3 = \frac{3 + \sqrt{5}}{2}, \qquad \lambda_4 = \frac{5 + \sqrt{5}}{2}. \tag{78}$$

Their approximate values are 0.381966011, 1.381966011, 2.618033989, and 3.618033989. The ratio targets are shown in Table 3.

Table 3: P5 frozen spectral-ratio targets relative to the lowest positive mode.

Ratio	R_{21}	R_{31}	R_{41}
S law, $\alpha = 1$	3.618033989	6.854101966	9.472135955
I law, $\alpha = 1/2$	1.902113033	2.618033989	3.077683537

For the general law $\omega_j = C\lambda_j^\alpha$, define

$$x_j = \ln \frac{\lambda_j}{\lambda_1}, \qquad y_j = \ln \frac{\widehat{\omega}_j}{\widehat{\omega}_1}. \tag{79}$$

A zero-intercept least-squares estimator is

$$\widehat{\alpha} = \frac{\sum_{j>1} x_j y_j}{\sum_{j>1} x_j^2}. \tag{80}$$

The overall calibration constant cancels. P5 is therefore stronger than P3 as a controlled discriminator because it provides three independent nontrivial ratio constraints rather than one.

12.4 EW-0 and implementation status

EW-0 is the frozen event-window synchronization protocol. Reconfiguration events are timestamped, settling windows are excluded or separately characterized, and S/I analysis windows use matched timing conventions so that switching transients are not mistaken for stationary spectral-law differences.

The intended mixed-model stack is:

- OPAx197 vendor operational-amplifier model;
- ADG333A vendor/ADI switch model;
- ADG1412 vendor switch model;
- datasheet-bounded behavioral LM319 comparator model.

The hardware checkpoint chain is

$$\text{H0-PASS} \to \text{HW-0} \to \text{POSTDR0-H2C-PARTIAL} \to \text{XR-0}. \tag{81}$$

XR-0 means SPICE execution-ready. The full vendor/behavioral transient remains unexecuted in the frozen record.

12.5 Evidence ladder

The current status is summarized in Table 4.

The hardware program is therefore a controlled test of whether finite analog/emulator systems can realize and discriminate declared operator laws. It is not presently a direct experimental test of the physical selector problem.

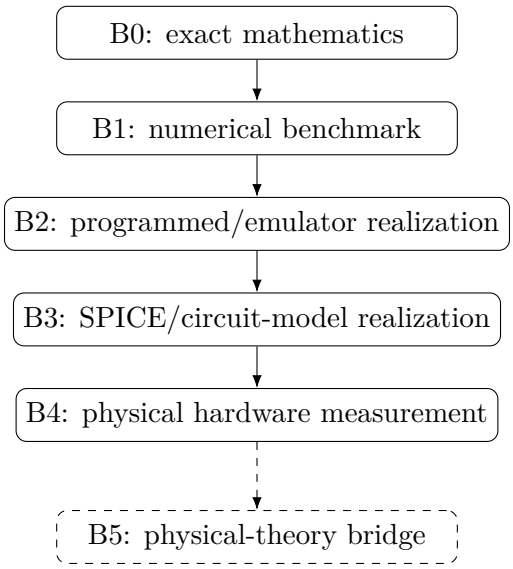


Figure 3: Benchmark evidence ladder. Even perfect implementation agreement at B4 does not by itself supply the B5 physical selector.

Table 4: Benchmark and hardware status.

Object	**Evidence level**	**Status**
Application A0	Numerical inverse benchmark	Completed
P3	Exact benchmark	Completed mathematically
P5	Exact controlled discriminator	Completed mathematically
Six-state S/I system	Programmed/emulator construction	Defined
EW-0	Measurement protocol	Frozen
Hardware package	Implementation specification	Frozen
XR-0	SPICE readiness	Execution-ready
Vendor/behavioral transient	Circuit-model execution	Not executed
Physical hardware measurement	Experimental	Not established
Physical-selector test	Physical-theory bridge	Open

13 Falsifiability and Empirical Access

Because HSMT presently admits multiple mathematical realizations, falsifiability must be stated at the correct logical level. A realization-level prediction may fail without falsifying every possible realization of the abstract framework. Conversely, a universal structural consequence, if one is identified, may constrain the framework itself.

Legitimate empirical targets include:

- failure of a bridge-specific scaling relation;
- failure of an emulator or hardware implementation to reproduce an exact benchmark within predeclared uncertainty;
- observational exclusion of a proposed physical selector;
- failure of a dimensional bridge to reproduce an operational dimension measurement;
- inconsistency of a proposed causal bridge with relativistic signaling behavior;
- incompatibility of a proposed universality mechanism with observed common spacetime structure.

The benchmark hierarchy in Fig. 3 therefore distinguishes implementation falsification from ontological falsification. Successful realization of the P3/P5 laws validates the implementation of those laws; it does not by itself establish that either is fundamental in nature.

14 Candidate Selection Principle: Maximal Collective Relaxation

14.1 Scientific status and relationship to prior optimization literature

The original graded core does not contain a mechanism selecting its physical generator. We therefore study a *separate, defeasible candidate* rather than silently adding a new axiom to the frozen core. The maximal collective-relaxation selection principle (MCRSP-1) proposes that, in a natural system with physically adaptive conductances, fixed connected graph topology, fixed total positive conductance, and a conserved collective mode, the persistent selected interaction pattern maximizes the slowest nonzero relaxation rate. The mathematical optimization of mixing rates for continuous-time graph Markov processes is established in the literature [1, 2]. The speculative proposition here is *physical spontaneous selection*; neither the general optimization objective nor the path-graph algebra is claimed as a new physical law or a novel graph-theory invention.

Hypothesis 14.1 (MCRSP-1: conditional physical-selection proposal)**.** *Within a clearly specified physical class of independently adaptive, connected, weighted networks with conserved total conductance W and uniform response rates, the stable selected weights maximize $\lambda_2(L(w))$, the smallest positive eigenvalue of the weighted graph Laplacian, over the physically accessible set of weights.*

The word *independently* matters: the adaptation equations, conservation rule, topology, and accessibility must follow from physical characterization rather than be retrofitted to the proposed optimum. MCRSP-1 is not assumed to apply to all closed quantum systems or to all HSMT sectors. In particular, relaxation is not a property of an arbitrary unitary evolution.

14.2 Variational formulation

For the path P_n with $n \geq 2$ vertices and edge conductances $w_i > 0$, define

$$L(w) = \sum_{i=1}^{n-1} w_i (e_i - e_{i+1})(e_i - e_{i+1})^{\mathsf{T}}, \qquad \sum_{i=1}^{n-1} w_i = W > 0. \tag{82}$$

The constant vector is the unique zero eigenvector. For the formal diffusion equation $\dot{p} = -L(w)p$, λ_2 controls the slowest nonstationary exponential decay. This parameter t is the time of the *declared diffusion model*, not an HSMT-derived clock. The conditional selection problem is

$$\underset{w_i>0,\ \sum_i w_i = W}{\text{maximize}} \quad \lambda_2(L(w)). \tag{83}$$

Theorem 14.2 (Optimal path weights)**.** *For the weighted n-vertex path with $n \geq 2$ and fixed sum of positive edge weights W, the unique maximizer of λ_2 is*

$$w_i^* = \frac{6W}{n(n^2-1)} i(n-i), \qquad 1 \leq i \leq n-1, \tag{84}$$

with optimum

$$\lambda_2(L(w^*)) = \frac{12W}{n(n^2-1)}. \tag{85}$$

Proof. Let $x_i = i - (n+1)/2$. Then $x \perp \mathbf{1}$, $x_{i+1} - x_i = 1$, and $\sum_i x_i^2 = n(n^2-1)/12$. The Rayleigh principle yields

$$\lambda_2(L(w)) \leq \frac{x^{\mathsf{T}} L(w) x}{x^{\mathsf{T}} x} = \frac{W}{n(n^2-1)/12}. \tag{86}$$

Write $c = 6W/[n(n^2-1)]$ and set $w_i = ci(n-i)$, with $w_0 = w_n = 0$. Since $(Lx)_i = w_{i-1} - w_i$, direct subtraction gives $Lx = 2cx$. The strictly increasing vector x has exactly one sign change (allowing a central zero); the standard oscillation ordering for a positive Jacobi/path Laplacian places it in the first nonconstant eigenspace. Thus $\lambda_2 = 2c$ and the bound is attained. If another choice attains the same bound, x minimizes the Rayleigh quotient on $\mathbf{1}^{\perp}$ and hence satisfies $L(w)x = \lambda_2 x$. The first-row equation fixes w_1, and subsequent rows recursively fix every w_i, recovering $w_i = ci(n-i)$. The maximizer is therefore unique. □

14.3 Exact finite spectra and an independent discriminator

For $n = 3$ and $W = 2$, the optimum is $w_1 = w_2 = 1$, with $\operatorname{spec} L = \{0, 1, 3\}$. For $n = 5$ and $W = 4$, one obtains

$$(w_1, w_2, w_3, w_4) = (0.8, 1.2, 1.2, 0.8), \qquad \operatorname{spec} L = \{0, 0.4, 1.2, 2.4, 4.0\}. \tag{87}$$

The positive eigenvalue ratios are $1 : 3 : 6 : 10$. By contrast, unit weights on all four edges give eigenvalues $2 - 2\cos(k\pi/5)$, $k = 0, \ldots, 4$, and a different positive spectrum. Such ratios are predictions of the *specified matrices*. If a programmable emulator is built with weights w^*, observing those ratios tests implementation, not spontaneous adaptation.

For the three-site family $w_1 = 1 + \varepsilon$, $w_2 = 1 - \varepsilon$, $|\varepsilon| < 1$, the positive gap is

$$\lambda_2 = 2 - \sqrt{1 + 3\varepsilon^2} = 1 - \frac{3}{2}\varepsilon^2 + O(\varepsilon^4). \tag{88}$$

Thus small asymmetric perturbations reduce the gap quadratically; this follows mathematically without

establishing an evolution law driving ε to zero.

14.4 Residual freedom: optimizing a grade block is not selecting physics

Proposition 14.3 (Internal-generator nonselection)**.** *Let L^* be the five-site optimizer above and let B be an otherwise free self-adjoint operator on $\mathbb{C}^2$. The family*

$$H_B = L^* \otimes I_2 + I_5 \otimes B \tag{89}$$

has the same selected grade Laplacian and the same $\lambda_2(L^*)$, but it need not have unitarily equivalent full generators.

Proof. Take $B_0 = 0$ and $B_1 = \mu\sigma_z$ with $\mu \neq 0$. The first operator has each L^* eigenvalue twice; the second has eigenvalues $\lambda_j \pm \mu$. For generic nonzero μ the multisets differ. Both operators are self-adjoint and generate norm-preserving one-parameter unitary groups. Their collective Laplacian is identical. □

This proposition blocks the inference that selecting a Laplacian determines HSMT's physical Hamiltonian, its internal fibres, its grade domain, or physical spacetime. Extending MCRSP-1 to a general noncommutative generator would require an additional physically justified objective. No such extension has been derived.

14.5 Experimental test and falsification protocol

Three noninterchangeable tests. *Mathematical optimization* verifies the extremizer of a declared objective; *engineered implementation* measures a system programmed with the proposed weights; *spontaneous physical selection* asks whether independently determined microscopic adaptation dynamics converge to those weights without being programmed to do so. Only the third test supplies direct evidence for the conjectured natural selection mechanism; even success there does not identify a fundamental HSMT law.

A genuine selection test must use a physical system whose adaptive rule and fixed-resource condition are independently characterized, rather than a network whose weights are manually programmed to match Eq. (84). Pre-specify the graph, equal node-response factors, conservation precision, observation horizon, admissible weight range, measurement uncertainties, and a quantitative criterion for convergence to a persistent configuration. For a qualifying five-site system with $W = 4$, MCRSP-1 predicts $w = (0.8, 1.2, 1.2, 0.8)$ up to the predeclared tolerances, with independently measurable relaxation spectrum as a cross-check. A reproducibly stable alternative pattern while the stipulated premises hold would reject MCRSP-1 *for that domain*. An engineered circuit reproducing Eq. (87) does not test spontaneous physical selection. No independent experiment of this kind is reported in the present manuscript.

14.6 Connection to the physical-selection ledger

The proposal has three distinct evidential layers: (i) a proven restricted graph optimization; (ii) an unverified hypothesis that certain natural adaptive networks spontaneously find its optimizer; (iii) an unestablished correspondence between such a network and HSMT's underlying dynamics. A positive result at (ii) would not prove (iii). In particular, MCRSP-1 does not derive $N = 4$, a Gaussian width, a physical arrow of time, Einstein gravity, Standard Model degrees of freedom, or a unique realized sector. The foundational selector remains open.

15 The Open Physical Selector Problem

The remaining physical ledger is finite and explicit.

Open Problem 15.1 (Physical selector). Which mathematical realization corresponds to the physical world?

Open Problem 15.2 (Physical spacetime bridge). What operational principle gives the abstract graded operator structure physical spacetime meaning?

Open Problem 15.3 (Physical dimension). What licenses identifying one mathematical dimensional diagnostic with observed physical dimension?

Open Problem 15.4 (Causal bridge). What establishes operational Lorentzian causal structure rather than only mathematical propagation?

Open Problem 15.5 (Multi-sector universality). Why should independently emergent sectors share one physical metric, dimensionality, causal cone, or spacetime realization?

Additional open items include the physical generator, the unique physical origin of the Gaussian kernel where not model-assumed, a possible grade-dependent width/separation law, packet-frame completeness, and empirical selection among admissible realizations.

15.1 Concise physical-completion ledger

For clarity to readers encountering HSMT for the first time, the missing physical identifications can be organized into four groups:

(i) **Selection:** an independently justified physical generator and realized P/Q sector, including a physical origin of coupling profiles;

(ii) **Spacetime:** operational time calibration, dimension, Lorentzian causal structure, and agreement across independently emergent sectors;

(iii) **Measurement:** a physically calibrated localization/POVM bridge, with packet-frame completeness established where needed;

(iv) **Discrimination:** empirical data that identify a physical realization rather than simply verifying an engineered emulator.

The complete itemized ledger appears in the appendix. No item in this summary is considered solved by the candidate principle MCRSP-1.

The MC-1 role is therefore an *envelope*: the function a genuine selector must perform has been identified, but no validated object occupies it. The conditional MCRSP-1 proposal in Section 14 optimizes a restricted collective Laplacian and demonstrably does not select the full generator.

16 Discussion

The mature HSMT program illustrates a distinction that is broadly relevant to emergence-based foundational theories: producing effective mathematical structure is not the same as deriving physical ontology. A single framework may admit effective sectors, memory kernels, spectral flows, running diagnostics, transport regimes, conserved records, and informationally complete measurement structures while still leaving open what those structures mean physically.

This distinction is particularly important when spacetime itself is intended to be emergent rather than assumed. The temptation is to rename a mathematically convenient object after the desired physical concept. HSMT's maturation process instead treats such identifications as separate claims requiring separate support.

The positive results are substantial. Exact projection theory supplies controlled effective dynamics. Spectral inversion can be benchmarked analytically and numerically. Scale-dependent diagnostics can be defined without presupposing a unique dimension concept. Internal measurement structure and persistent records can be analyzed. Finite graph systems provide exact discriminators for distinct spectral laws. These are useful mathematical and computational assets independent of whether the final physical bridge has yet been found.

The negative results are equally constructive. NI-0 and NI-1F rule out broad classes of internally generated selectors. MS-C0 blocks a mode-count-to-dimension shortcut. PB-0 shows that exact effective reduction is not sector selection. PB0-N1 through PB0-N7 separate invariant characteristic information from grade origin, Lorentzian causality, light-speed normalization, and dimension recovery. OMP-3F records a genuine valid stop rather than forcing a premature completeness claim.

The resulting framework is more constrained than earlier developmental versions, but scientifically cleaner. The next decisive advance must come from experiment, an independently motivated microscopic principle, a cross-theory operational bridge, or another genuinely external source of information.

Historical cosmological ideas—including redshift, supernova dimming, CMB shell-boundary interpretations, and fractional-dimensional progression—remain possible research hypotheses rather than consequences of the frozen framework. The present paper does not use them as evidence for physical closure.

17 Conclusion

Hierarchical Shell Manifold Theory has reached a mature internal checkpoint. Its frozen core consists of a graded Hilbert-state and operator architecture capable of supporting controlled self-adjoint realizations, exact projection, effective operators and memory, spectral reconstruction, dimensional diagnostics, measurement structures, record-persistence conditions, and exact and numerical benchmark systems.

The same maturation program establishes equally important limits. The present premise set does not derive physical spacetime, observed physical dimension, Lorentzian causality, a uniquely realized sector, operational physical time, or universal geometry across independently emergent sectors. These open questions cannot be closed by terminology, reparameterization, or reusing the present premises in disguised form.

HSMT-FM0 therefore marks a principled internal derivational boundary rather than the completion of physical theory. The next phase is external-facing: execute the controlled benchmarks already specified, confront candidate bridges with observation, and search for independently motivated principles capable of selecting among mathematically admissible realizations.

In this form, HSMT is an extensible framework with explicitly testable conditional realizations, including the separately identified MCRSP-1 hypothesis, while framework-level physical selection remains open.

A Notation, Status Vocabulary, and Closure Terminology

Symbol/term	Meaning
$\mathcal{H}$	Hilbert-state arena.
P_N	Orthogonal grade projector.
N	Grade label; not physical dimension by definition.
D	Reference operator.
H	Generic self-adjoint dynamical/spectral operator when a specific evolution realization is used.
P, Q	Chosen effective/complement projections, $Q = I - P$.
τ	Mathematical evolution parameter; not automatically physical time.
$Q_N(\ell)$	Declared scale-dependent diagnostic associated with grade N.
$d_N(\ell)$	Local logarithmic scaling exponent of Q_N.
$d_{\mathrm{geom}}, d_{\mathrm{res}}, d_{\mathrm{spec}}, d_{\mathrm{dyn}}$	Distinct dimension notions; equality not assumed.
β_Ω	Historical inverse-square branch exponent.
β_{fr}	WKB frame-density exponent.
ν_B	Boundary/Bessel index in the conditional inverse-square branch.
HSMT-FM0	Terminal internal checkpoint: current-premise derivation complete to boundary, physical completion open.

Claim-status and closure terminology are defined in Section 10 and the opening methodological section. The distinction between no-go, non-selection, and valid stop must be maintained in any later extension.

B Operator Domains and Self-Adjointness Details

Theorem 4.1 is the principal operator-domain result required for the bounded-hopping constructions used in the main text. More generally, if a symmetric perturbation V is D_0-bounded with relative bound $a < 1$, i.e.

$$\|V\psi\| \le a\|D_0\psi\| + b\|\psi\|, \qquad \psi \in \mathrm{Dom}(D_0), \tag{90}$$

then Kato–Rellich yields self-adjointness of $D_0 + V$ on $\mathrm{Dom}(D_0)$. Bounded hopping is the special case $a = 0$.

The theorem concerns the declared realization. It does not select D_0, V, the grade couplings, or a physical generator.

C Exact Feshbach Reduction and Memory Kernel

For completeness, the block operator $z - H$ is

$$z - H = \begin{pmatrix} z - H_{PP} & -H_{PQ} \\ -H_{QP} & z - H_{QQ} \end{pmatrix}. \tag{91}$$

Eliminating the Q block by block Gaussian elimination gives the Schur complement (17) and projected resolvent (18). The time-domain kernel (26) follows by solving the complementary equation and substituting (23) into the P equation.

The exact initial-Q source in (25) is important: an initially occupied eliminated sector cannot generally be represented only by a homogeneous memory convolution.

D Spectral Inversion and Conditional Asymptotics

D.1 Stieltjes consistency checks

For a finite positive measure μ with convention (30), $m(z)$ is analytic on $\mathbb{C} \setminus \operatorname{supp} \mu$, satisfies $\operatorname{Im} m(z) < 0$ for $\operatorname{Im} z > 0$ unless μ is trivial, and obeys

$$m(z) = \frac{\mu(\mathbb{R})}{z} + O(z^{-2}) \tag{92}$$

when the required moments exist. For a normalized contact measure, $zm(z) \to 1$.

D.2 A0 noise response

The approximately linear transfer from multiplicative complex noise in G to relative error in Σ over the tested range is empirical to the frozen benchmark and should not be extrapolated without a separate conditioning analysis.

D.3 Historical chain

Equation (43) is retained exactly at conditional status. If the original source equations connecting A, M, μ, Σ, p, and E are recovered, they should be inserted here. In their absence the chain is not to be reverse-engineered into a new derivation during synthesis.

E Dimensional Diagnostics, SRB Counts, and MS-C0

For any positive differentiable $Q(\ell)$,

$$\frac{\mathrm{d} \ln Q}{\mathrm{d} \ln \ell} = \frac{\ell Q'(\ell)}{Q(\ell)}. \tag{93}$$

If $Q(\ell) = C\ell^d$, this equals d. If $Q(M) \sim CM^k$, the same identity gives a count exponent tending to k. Neither identity supplies physical dimensional meaning without an operational bridge.

The frozen SRB record also includes the onset datum $q_I(4) = 2$, the qualified Clifford-type identity for Δ_m, and the distinction between positive Gram structure, elementwise positivity, and strict overlap. Exact source formulas for q_I and the S^1 Fourier counterexample should be restored if available in the archival record.

F No-Go and Boundary Ledger

Item	Status	Frozen content
NI-0	No-go	Monotone positive-semidefinite q-couplings do not generate the required interior restoring minimum in the declared class.
NI-1F	Theorem/no-go	Gaussian measured-pullback functional is monotone; no canonical interior selector in the declared class.
B3	Non-selection	$V_{\mathrm{gr-rel}}$ not selected.
B5	Non-selection	Seed/absolute grade origin not derived.

Item	Status	Frozen content
B7	Non-selection	$W(\rho)$ not selected.
Transport	No-go against overreach	Locality/dephasing/clustering alone do not imply diffusion.
S^1 positivity	Counterexample	Positive Gram, elementwise positivity, and strict overlap are not equivalent.
OMP-3F	Valid stop	Packet-frame completeness remains undetermined.
MS-C0	No-go against conflation	Mode-count exponent is not physical dimension without an operational bridge.
BEGC	Conditional mathematical result	Exposed-state/universal-law family has lower finite-resolution complexity than generic pairwise data; not a physical selector.
PB-0	Principled non-selection	Chosen P/Q can be analyzed; physical P/Q is not selected.
PB0-N1–N7	Boundary sequence	Characteristic, grade-origin, causality, normalization, and dimension shortcuts excluded.
MC-1	Open envelope	Selector role identified; no occupant.
HSMT-FM0	Terminal internal boundary	Current-premise internal derivation complete to boundary; physical completion open.

G POVM, Packets, and Records

The publication-level summary retained from the frozen PC/OMP sequence is:

1. canonical POVM relative to the declared internal-fibre construction;
2. informational completeness on the internal fibre;
3. no canonical localized packet selected by the current internal structure;
4. packet-frame completeness unresolved at OMP-3F;
5. Q_N is a scalar diagnostic only;
6. RS-0 record persistence under (56) with the appropriate operator-domain conditions.

If the exact canonical POVM formula, PC wording, and OMP-3F source statement are recovered, they should replace this conservative archival summary in a subsequent publication revision.

H P3, P5, S+C, and Spectral-Law Benchmarks

H.1 P3 characteristic polynomial

For (70), direct evaluation gives

$$\det(K - \lambda I) = -\lambda(\lambda - 1)(\lambda - 3), \tag{94}$$

confirming (71).

H.2 General power-law discriminator

For $\omega = C\lambda^{\alpha}$,

$$\ln \frac{\omega_j}{\omega_1} = \alpha \ln \frac{\lambda_j}{\lambda_1}, \tag{95}$$

which motivates (80). The ratio test is calibration independent but not implementation-error independent.

H.3 S+C

S+C remains separate from P3/P5. The former is a minimal nonfundamental relational/refinement benchmark; the latter are controlled spectral dynamical benchmarks on a fixed finite operator. Neither supplies the physical selector.

I Hardware Architecture and Execution Status

The ideal state matrices corresponding to (72) and (74) are

$$A_S = \begin{pmatrix} 0 & gK \\ -gK & 0 \end{pmatrix}, \qquad A_I = \begin{pmatrix} 0 & gI \\ -gK & 0 \end{pmatrix}. \tag{96}$$

A physical or circuit realization has

$$A_{\text{real}} = A_{\text{target}} + \Delta A, \tag{97}$$

where ΔA collects finite gain/bandwidth, component tolerances, switch resistance/charge injection, comparator delay/hysteresis, slew rate, offsets, parasitics, supply limits, and switching transients.

A future result should predeclare ratio/exponent tolerances before inspecting the experimental outcome. A generic analysis statistic is

$$\epsilon_{\text{ratio}}^{(X)} = \left[\frac{1}{m} \sum_{j=2}^{m+1} \left(\frac{\widehat{R}_{j1} - R_{j1}^{(X)}}{R_{j1}^{(X)}} \right)^2 \right]^{1/2}, \tag{98}$$

with $X \in \{S, I\}$. This is an analysis proposal, not a new physical axiom; any exact frozen tolerance budget should supersede it.

The present archival status remains: execution-ready SPICE package, full vendor/behavioral transient unexecuted, physical bench measurement not established.

J Frozen Checkpoint Ledger

The mature checkpoint record is grouped by scientific role.

Checkpoint family	Scientific role
SRB-0–SRB-5	Structural/spectral audit: onset, mode counts, Δ_m identity, positivity qualifications, S^1 counterexample.
PCS / PCS-RP0	Finite-width feature map; valid execution / partial diagnostic failure; no RP continuation.
OMP sequence	Packet/frame program; OMP-3F valid stop.

Checkpoint family	Scientific role
PC-0–PC-4	Canonical POVM, internal-fibre IC, no canonical localized packet, Q_N scalar diagnostic.
FC-0	Minimal core $(\mathcal{H}, \{P_N\}, D)$ frozen.
PG-0	Physical generator not selected.
DCP-0	Norm-preserving closed evolution adopted.
PI-0	Evolution/time/causality firewall frozen.
B3/B5/B7	Post-PI0 non-selection results.
POSTFS0-SA1 / RS-0 / PSB1-B	Structural, record-persistence, and ring-exchange analogue branch.
H0-PASS / HW-0 / H2C-PARTIAL / XR-0 / EW-0	Exact benchmark through execution-ready hardware branch.
NI-0 / NI-1 / NI-1F	Monotone-coupling and Gaussian measured-pullback vacuum-selection no-go lineage.
S+C	Minimal nonfundamental refinement/cycle benchmark.
BEGC-0/1	No-smuggling / exposed-state universal-law complexity branch.
PB-0	Physical projector branch closed at principled non-selection.
PB0-N1–N7	Characteristic/causal boundary sequence.
HSMT-FM0	Terminal internal maturation checkpoint.

HSMT-FM0 is the terminal checkpoint of the present internal-development phase. It does not denote physical completion.

K Conditional Bridges, MC-1, and External Reopening

The MC-1 selector envelope is defined functionally: an occupant must discriminate among mathematically admissible realizations and connect the selected structure to operational observables through independently justified input. No current object occupies the envelope.

A future branch may reopen foundational derivation only through genuinely new empirical, microscopic, mathematical, or cross-theory information as described in Section 11. A conclusion-driven insertion does not count as a new bridge.

L Superseded Developmental Material and Provenance

The following classes of earlier developmental language are not live claims of the mature framework:

- literal ordinary concentric spatial-shell ontology as a foundational primitive;
- direct identification $N =$ physical dimension;
- direct derivation of physical $3 + 1$ from the choice $N = 4$;
- automatic elliptic-to-causal inference;
- unqualified equality $d_{\mathrm{geom}} = d_{\mathrm{res}} = d_{\mathrm{spec}} = d_{\mathrm{dyn}}$;
- treating coordinate reparameterization as dynamical selection;

- claiming locality/dephasing alone forces diffusion;
- positivity claims contradicted by the frozen S^1 counterexample;
- unverified screening, vacuum-selection, or cosmological claims promoted beyond their frozen status.

These items are retained here only to make the maturation boundary transparent.

Summary Status Tables

Table 8: Major positive results and their firewalls.

Result	**Status**	**Physical firewall**
Bounded-hopping self-adjointness	Theorem	Does not select physical generator
Exact Feshbach reduction	Exact theorem	Assumes P; does not select physical sector
Memory kernel	Exact derived result	Not an independent arrow of time
Scalar $G \to \Sigma$	Exact model identity	Does not select physical spectral measure
Semicircle A0	Analytic/numerical benchmark	Engine verification, not physical validation
WKB density map	Derived given W	Does not select W or identify frame index with grade
$Q_I, Q_H \sim M^3/6$	Structural asymptotic	Count exponent is not physical dimension
Canonical POVM / internal-fibre IC	Derived in frozen construction	Not physical position or spacetime localization
RS-0	Conservation condition	Not physical time or arrow
P3/P5	Exact controlled benchmarks	Not physical HSMT validation

Table 9: Open physical ledger at HSMT-FM0.

Problem	**Status**
Physical selector	OPEN
Physical P/Q decomposition	OPEN
Physical spacetime bridge	OPEN
Physical dimension bridge	OPEN
Lorentzian causal bridge	OPEN
Operational physical-time bridge	OPEN
Multi-sector universality	OPEN
Physical generator	OPEN
Unique physical Gaussian-kernel origin	OPEN where not model-assumed
Grade-dependent width/separation law	OPEN
Packet-frame completeness	OPEN / VALID STOP
Empirical selection among admissible realizations	OPEN

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